



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 3

SHEET

1

of

2

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD **ROTARY**

CO-ORDINATES

E 829001.64

N 827502.52

TASK ORDER NO.

GE/2022/08.11B

MACHINE

EW06

DATE

19.12.2022 to 21.12.2022

FLUSHING MEDIUM

WATER

ORIENTATION **VERTICAL**

GROUND LEVEL

+178.57 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
19.12.2022	PW													
1														Brownish yellow (10YR 6/6), slightly clayey sandy SILT with some angular to subangular fine to medium gravel. (FILL)
2			60	40										Light yellowish brown (10YR 6/4), angular to subangular, coarse GRAVEL of rock fragments. (FILL)
3			60	100									IV	Weak, pale yellow (2.5Y 7/4) dappled yellowish red, highly decomposed, coarse ash TUFF. (Angular, slightly silty sandy fine to medium GRAVEL of highly decomposed tuff fragments)
4			60	100									IV	Weak, brownish yellow (10YR 6/8) dappled dark grey, highly decomposed, TUFF BRECCIA. (Angular, slightly silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
5			60	100										
6			60	99										
7			60	100									V	Extremely weak to very weak, light yellowish brown (2.5Y 6/4) and brownish yellow (10YR 6/8), completely decomposed, coarse ash TUFF. (Slightly clayey silty fine to coarse SAND with much angular fine to medium gravel)
8			60	88										
9			60	60										
10			60	60										

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

28.12.2022

CHECKED

R. Chu

DATE

03.01.2023

REMARKS

- An inspection pit was excavated to 1.10m deep by hand tools.
- Constant head permeability tests were carried out at sections from 3.40m to 4.90m and from 8.60m to 10.10m.
- A standpipe was installed to 2.00m.
- A piezometer was installed with tip at 14.50m.

DRiLTECH DRILLHOLE RECORD CONTRACT NO. GE/2022/08 HOLE NO. DH 3 SHEET 2 of 2									
PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk									
METHOD ROTARY					CO-ORDINATES E 829001.64 N 827502.52		TASK ORDER NO. GE/2022/08.11B		
MACHINE EW06							DATE 19.12.2022 to 21.12.2022		
FLUSHING MEDIUM WATER					ORIENTATION VERTICAL		GROUND LEVEL +178.57 mPD		
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return %	TCR %	SCR %	RQD %	Fracture Index	Tests	Description
20.12.2022 21.12.2022		1800 7.94 at 0800					15.0		As sheet 1 of 2.
11			60	74	18	9	NI	T2-101	Moderately weak to moderately strong, light brownish grey, moderately decomposed, coarse ash TUFF. (CORESTONE)
12			60	96	6	0	NR	T2-101	Weak to moderately weak, light yellowish brown (2.5Y 6/4), highly decomposed, coarse ash TUFF. (Angular, slightly sandy clayey silty fine to coarse GRAVEL of highly decomposed tuff fragments)
13			60	83	0	0	NR	T2-101	From 11.15m to 11.51m: No recovery, assumed to be completely decomposed TUFF.
14	HW 14.00m		60	0			NI	T2-101	Weak to moderately weak, light grey (10YR 7/1) mottled light brownish grey, highly decomposed, TUFF BRECCIA. (Angular, fine to coarse GRAVEL and occasional cobble of highly decomposed tuff breccia fragments)
21.12.2022		9.89 at 1800	60	100	19	19	>20	T2-101	From 13.39m to 13.60m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
16									Weak to moderately weak, light yellowish brown (2.5Y 6/4), highly decomposed, TUFF BRECCIA. (Angular, slightly silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
17									Moderately weak to moderately strong, light greyish brown, moderately decomposed, TUFF BRECCIA. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, dipping at 15° to 25°, 25° to 35° and 65° to 75°.
18									End of hole at 15.01 m.
19									
20									

SMALL DISTURBED SAMPLE
 LARGE DISTURBED SAMPLE
 U76 SAMPLE
 PISTON SAMPLE (76mm)
 MAZIER SAMPLE
 SPT LINER SAMPLE
 WATER SAMPLE
 U100 SAMPLE

STANDARD PENETRATION TEST
 IN-SITU VANE SHEAR TEST
 PACKER TEST
 PERMEABILITY TEST
 PRESSUREMETER TEST
 BOREHOLE TELEVIEWER
 PIEZOMETER TIP
 STANDPIPE TIP

LOGGED C. Chan

DATE 28.12.2022

CHECKED R. Chu

DATE 03.01.2023

REMARKS

DRiLTECH

DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO. **DH 4**

SHEET **1** of **3**

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD **ROTARY**

CO-ORDINATES

TASK ORDER NO. **GE/2022/08.11B**

MACHINE **SD44**

E 829015.24
N 827503.60

DATE **17.12.2022** to **22.12.2022**

FLUSHING MEDIUM **WATER**

ORIENTATION **VERTICAL**

GROUND LEVEL **+176.28 mPD**

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
17.12.2022	SW													Brown (7.5YR 5/4) mottled yellowish red, slightly clayey SILT with much angular to subangular fine to coarse gravel. (FILL)
1	SW 1.50m		85	76					1	0.00	0.00			
2	PW		80	100					2	0.45	1.00			Greyish brown (10YR 5/2) and reddish yellow (7.5YR 8/6), angular to subangular, clayey silty sandy fine to coarse GRAVEL and much cobble of rock fragments. (FILL)
3									3	0.95	1.30			Very stiff, reddish yellow (7.5YR 8/6) mottled brownish yellow, clayey sandy SILT with some angular to subangular fine to coarse gravel. (COLLUVIUM)
4			80	83					4	1.50				
5		1.17 at 1800							5	2.50				
6		4.49 at 0800							6	2.60				
7									7	2.70				
8									8	3.00				
9									9	3.30				
10									10	3.50				
11									11	3.50				
12									12	4.50				
13									13	4.60				
14									14	4.88				
15									15	5.50				
16									16	6.50				
17									17	6.82				
18									18	7.50				
19									19	8.25				
20									20	8.35				
21									21	8.35				
22									22	9.00				
23									23	9.00				
24									24	9.51				
25									25	9.51				
26									26	9.80				
27									27	9.90				
28									28	9.90				
29									29	9.90				
30									30	9.90				
31									31	9.90				
32									32	9.90				
33									33	9.90				
34									34	9.90				
35									35	9.90				
36									36	9.90				
37									37	9.90				
38									38	9.90				
39									39	9.90				
40									40	9.90				
41									41	9.90				
42									42	9.90				
43									43	9.90				
44									44	9.90				
45									45	9.90				
46									46	9.90				
47									47	9.90				
48									48	9.90				
49									49	9.90				
50									50	9.90				
51									51	9.90				
52									52	9.90				
53									53	9.90				
54									54	9.90				
55									55	9.90				
56									56	9.90				
57									57	9.90				
58									58	9.90				
59									59	9.90				
60									60	9.90				
61									61	9.90				
62									62	9.90				
63									63	9.90				
64									64	9.90				
65									65	9.90				
66									66	9.90				
67									67	9.90				
68									68	9.90				
69									69	9.90				
70									70	9.90				
71									71	9.90				
72									72	9.90				
73									73	9.90				
74									74	9.90				
75									75	9.90				
76									76	9.90				
77									77	9.90				
78									78	9.90				
79									79	9.90				
80									80	9.90				
81									81	9.90				
82									82	9.90				
83									83	9.90				
84									84	9.90				
85									85	9.90				
86									86	9.90				
87									87	9.90				
88									88	9.90				
89									89	9.90				
90									90	9.90				
91									91	9.90				
92									92	9.90				
93									93	9.90				
94									94	9.90				
95									95	9.90				
96									96	9.90				
97									97	9.90				
98									98	9.90				
99									99	9.90				
100									100	9.90				

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE
- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED **C. Chan**
DATE **28.12.2022**
CHECKED **R. Chu**
DATE **03.01.2023**

REMARKS
1. An inspection pit was excavated to 1.00m deep by hand tools.
2. Constant head permeability tests were carried out at sections from 3.10m to 4.60m and from 16.10m to 17.60m.
3. A standpipe was installed to 4.50m.
4. A piezometer was installed with tip at 18.00m.

DRiL TECH

DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 4

SHEET

2

of

3

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD ROTARY

CO-ORDINATES

TASK ORDER NO.

GE/2022/08.11B

MACHINE SD44

E 829015.24

N 827503.60

DATE

17.12.2022

to

22.12.2022

FLUSHING MEDIUM WATER

ORIENTATION VERTICAL

GROUND LEVEL

+176.28 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
11		at 0800	80	100	0	0	>20		T2-101		10.45		III	clayey SILT) Moderately weak to moderately strong, light grey and light yellowish brown, moderately decomposed, TUFF BRECCIA. (CORESTONE)
12			80	82	45	23	12.7		T2-101	+165.08	11.20		IV	Weak to moderately weak, yellowish brown (10YR 5/8), highly decomposed, TUFF BRECCIA. (Angular, clayey silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
13			80	96	0	0	NI		T2-101	+164.78	11.50		V	From 11.50m to 11.73m and 13.10m to 13.30m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
14			80	64	0	0	NR		T2-101	+164.55	11.73		IV	
15			80	0					T2-101	+163.18	13.10		V	Extremely weak, pale olive (5Y 6/3), completely decomposed, TUFF BRECCIA. (Stiff, clayey sandy SILT)
16			80	86	0	0	NI		T2-101	+162.99	13.30		IV	Weak to moderately weak, olive grey 5Y 5/2), highly decomposed, TUFF BRECCIA. (Angular, clayey silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
17			80	82					T2-101	+162.64	13.64		V	From 14.38m to 14.50m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
18			80	82					T2-101	+161.90	14.38		IV	From 17.60m to 17.73m: Moderately strong and moderately decomposed. (CORESTONE)
19			80	60					T2-101	+161.78	14.50		IV	
20			80	100	15	0	NI		T2-101	+158.68	17.60		III	Moderately strong, light grey dappled light yellowish brown, moderately decomposed, TUFF BRECCIA. (CORESTONE)
21			80	96	63	63	7.4		T2-101	+158.55	17.73		IV	From 18.98m to 19.48m: Strong and slightly decomposed.
22			80	96	63	63	7.4		T2-101	+157.70	18.58		III	From 19.93m to 20.22m: No recovery, assumed to
23			80	96	63	63	7.4		T2-101	+157.30	18.98		II	
24			80	96	63	63	7.4		T2-101	+156.80	19.48		III	
25			80	96	63	63	7.4		T2-101	+156.35	19.93		III	

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

28.12.2022

CHECKED

R. Chu

DATE

03.01.2023

REMARKS

										DRILLHOLE RECORD CONTRACT NO. GE/2022/08										HOLE NO. DH 4	
										SHEET 3 of 3											
PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk																					
METHOD ROTARY										CO-ORDINATES E 829015.24 N 827503.60					TASK ORDER NO. GE/2022/08.11B						
MACHINE SD44															DATE 17.12.2022 to 22.12.2022						
FLUSHING MEDIUM WATER										ORIENTATION VERTICAL					GROUND LEVEL +176.28 mPD						
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return %	TCR %	SCR %	RQD %	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description							
21	HW 20.43m		80	65	57	42	NR		T2-101	+158.08	20.22		V	be completely decomposed TUFF BRECCIA. Strong, light grey dappled light yellowish brown and grey, slightly decomposed, TUFF BRECCIA. Joints are medium to widely spaced, occasionally closely spaced, rough planar and rough stepped, iron oxide stained, chlorite and calcite coated, dipping at 0° to 10°, 10° to 20° and subvertically from 24.66m to 24.76m. From 20.22m to 20.60m and 25.24m to 25.42m: Moderately strong to strong and moderately decomposed. From 25.14m to 25.24m: No recovery, assumed to be completely decomposed TUFF BRECCIA.							
			80	100	100	100	5.3		T2-101	+155.68	20.60		III								
			80	100	100	100			T2-101				II								
			80	100	100	100			T2-101												
			80	100	100	100			T2-101												
22	6.32 at 1800 17.15 at 0800		80	100	100	100	0.2		T2-101		22.15										
			80	100	100	100			T2-101		23.18										
23			80	100	100	100			T2-101		24.66										
			80	100	100	100			T2-101		25.06										
24			80	100	100	100			T2-101		25.14										
			80	100	100	100			T2-101		25.14										
25	17.58 at 1800		80	93	82	82	5.0		T2-101	+151.14	25.14		V								
			80	93	82	82	NR		T2-101	+151.04	25.24		III								
26			80	93	82	82	2.5		T2-101	+150.88	25.42		II								
			80	93	82	82			T2-101	+150.25	26.03										
27														End of hole at 26.03 m.							
28																					
29																					
30																					

 SMALL DISTURBED SAMPLE
 LARGE DISTURBED SAMPLE
 U70 SAMPLE
 PISTON SAMPLE (76mm)
 MAZIER SAMPLE
 SPT LINER SAMPLE
 WATER SAMPLE
 U100 SAMPLE

 STANDARD PENETRATION TEST
 IN-SITU VANE SHEAR TEST
 PACKER TEST
 PERMEABILITY TEST
 PRESSUREMETER TEST
 BOREHOLE TELEVIEWER
 PIEZOMETER TIP
 STANDPIPE TIP

LOGGED C. Chan
 DATE 28.12.2022
 CHECKED R. Chu
 DATE 03.01.2023

REMARKS



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 5

SHEET

1 of 3

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD **ROTARY**

CO-ORDINATES

TASK ORDER NO.

GE/2022/08.11B

MACHINE

SD44**E 829036.19****N 827484.26**

DATE

09.12.2022 to 14.12.2022

FLUSHING MEDIUM

WATERORIENTATION **VERTICAL**

GROUND LEVEL

+171.34 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
09.12.2022	SW													
1			80	76					1	+171.34	0.00			Pale brown (10YR 6/3), clayey sandy SILT with much angular to subangular fine to coarse gravel of rock fragments and with some rootlets. (FILL)
									T2-101	+170.84	0.50			Dark grey (10YR 4/1) mottled light brownish grey and yellowish brown, angular to subangular, COBBLE and much coarse gravel of rock fragments. (FILL)
2	SW 1.55m		80	67										
	PW								T2-101		1.55			
3			80	0						+169.04	2.30			Light grey (10YR 7/1) and light yellowish brown (10YR 6/4), angular to subangular, clayey silty sandy fine to coarse GRAVEL and much cobble of rock fragments. (COLLUVIUM)
4			80	95					2	+167.94	3.40			Yellow (10YR 8/8) dappled grey, angular to subangular, slightly clayey silty sandy fine to coarse GRAVEL of rock fragments. (COLLUVIUM)
5									3		3.40			
									4		4.40			
									5		4.50			
									6		4.60			
6			80	100					7		5.40			
	PW 5.90m								8	+165.44	5.90			
	HW		80	97					T2-101		6.65			Grey (10YR 6/1) mottled light brownish grey and yellowish brown, angular to subangular, medium to coarse GRAVEL and much cobble of rock fragments. (COLLUVIUM)
09.12.2022		0.42 at 1800								+164.58	6.75			
10.12.2022		1.72 at 0800												
7			80	93	36	22	14.3		T2-101		7.03		III	Moderately weak to moderately strong, grey and yellowish brown, moderately decomposed, coarse ash TUFF. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, dipping at 0° to 10°, 10° to 20° and 65° to 75°.
8										+163.26	8.08			
			80	100	83	72			T2-101		9.11		III	From 7.50m to 8.08m: Moderately weak. Moderately strong to strong, grey dappled yellowish brown and light olive green, moderately decomposed, coarse ash TUFF. Joints are closely to medium spaced, occasionally very closely spaced, rough planar and rough stepped, iron and manganese oxide stained, chlorite coated, dipping at 0° to 10°, 15° to 25°, 45° to 65° and 65° to 75°.
9														
10			80	100	100	100	3.6		T2-101	+161.34	10.00			From 11.40m to 11.60m, 15.60m to 15.73m and

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

20.12.2022

CHECKED

R. Chu

DATE

21.12.2022

REMARKS

- An inspection pit was excavated to 0.50m deep by hand tools.
- A standpipe was installed to 6.50m.
- A piezometer was installed with tip at 21.00m.



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO. **DH 5**SHEET **2** of **3**

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD		ROTARY		CO-ORDINATES		TASK ORDER NO.								
MACHINE		SD44		E 829036.19 N 827484.26		GE/2022/08.11B								
FLUSHING MEDIUM		WATER		ORIENTATION		VERTICAL								
GROUND LEVEL						+171.34 mPD								
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
11			80	85	57	53	NR		T2-101	10.63			III	16.98m to 17.20m: No recovery, assumed to be completely decomposed TUFF.
12			80	100	90	72	>20		T2-101	+159.94 +159.74	11.40 11.60		V	From 12.15m to 12.32m: Tuffaceous SILTSTONE.
13			80	100	90	72	5.4		T2-101	12.05	12.20		III	From 14.05m to 14.15m, 14.95m to 15.12m and 15.43m to 15.60m: Weak and highly decomposed. (Angular, slightly sandy fine to coarse GRAVEL of highly decomposed tuff fragments)
14			80	100	90	79	NI		T2-101	13.51			IV	From 14.15m to 14.80m, 15.73m to 16.41m and 16.62m to 16.98m: Strong and slightly decomposed.
15			80	94	68	64	6.3		T2-101	+157.29 +157.18	14.05 14.15		II	
16			80	66	21	0	NI		T2-101	+156.54 +156.39 +156.22	14.80 14.95 15.12		III	
17			80	100	89	81	2.9		T2-101	+155.91 +155.74 +155.61	15.35 15.60 15.73		IV	
18			80	100	89	81	8.8		T2-101	+154.93 +154.72	16.41 16.62		III	
19			80	42	42	42	NR		T2-101	+154.36 +154.14	16.98 17.20		V	
20			80	0					T2-101	17.20			IV	Weak, reddish yellow (5YR 6/6), highly decomposed, coarse ash TUFF. (Angular, slightly silty clayey sandy fine to coarse GRAVEL of highly decomposed tuff fragments)
			80	70	0	0	NI		T2-101	+153.04 +152.71 +152.57	18.30 18.63 18.77		IV	Weak to moderately weak, yellowish brown (10YR 5/4) mottled grey, highly decomposed, coarse ash TUFF. (Angular, clayey silty sandy fine to coarse GRAVEL and some cobble of highly decomposed tuff fragments)
			80	100	0	0	NI		T2-101	+152.17	19.17		V	
			80	84	15	15	>20		T2-101	19.17	19.36		III	From 18.63m to 18.77m: No recovery, assumed to be completely decomposed TUFF.
			80	84	15	15	>20		T2-101	20.00			III	Moderately weak to moderately strong, light

SMALL DISTURBED SAMPLE

LARGE DISTURBED SAMPLE

U78 SAMPLE

PISTON SAMPLE (76mm)

MAZIER SAMPLE

SPT LINER SAMPLE

WATER SAMPLE

U100 SAMPLE

STANDARD PENETRATION TEST

IN-SITU VANE SHEAR TEST

PACKER TEST

PERMEABILITY TEST

PRESSUREMETER TEST

BOREHOLE TELEVIEWER

PIEZOMETER TIP

STANDPIPE TIP

LOGGED **C. Chan**

DATE **20.12.2022**

CHECKED **R. Chu**

DATE **21.12.2022**

REMARKS



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 5

SHEET

3 of 3

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD ROTARY

CO-ORDINATES

TASK ORDER NO.

GE/2022/08.11B

MACHINE SD44

E 829036.19

N 827484.26

DATE

09.12.2022 to 14.12.2022

FLUSHING MEDIUM WATER

ORIENTATION VERTICAL

GROUND LEVEL

+171.34 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description	
21 13.12.2022 14.12.2022 22.73m		0.93 at 1800 8.60 at 0800	80	75			NR	▲	T2-101 16-20.20	+151.30 +151.14	20.04 20.20		III IV	brownish grey, moderately decomposed, coarse ash TUFF. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, clayey silt coated, dipping at 0° to 10°, 10° to 20° and 65° to 75°. From 20.04m to 20.20m: No recovery, assumed to be completely decomposed TUFF. Weak, pale brown (10YR 6/3), highly decomposed, coarse ash TUFF. (Angular, slightly silty sandy fine to coarse GRAVEL and some cobble of highly decomposed tuff fragments) Moderately weak to moderately strong, light olive grey stripped yellowish brown, moderately decomposed, SILTSTONE. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, dipping at 0° to 10°, 10° to 20°, 35° to 45° and 65° to 75°. From 21.41m to 21.84m: Weak to moderately weak and highly decomposed. (Angular, medium to coarse GRAVEL of highly decomposed siltstone fragments) From 22.29m to 22.43m: No recovery, assumed to be completely decomposed SILTSTONE. Strong to very strong, dark grey and bluish grey dappled pink and light grey, slightly decomposed, coarse ash TUFF. Joints are very widely spaced, rough planar, calcite and chlorite coated, dipping at 5° to 15°. From 22.43m to 22.73m: Moderately strong and moderately decomposed. From 22.43m to 22.73m and 24.96m to 25.55m: Tuffaceous SILTSTONE. From 24.64m to 24.74m: With some quartz veins (<10mm thick).	
			80	100	92	0	>20			T2-101 11-20.80 20.90	+150.44	20.90			III
			80	100	23	0	NI			T2-101 21.28	+149.93	21.41			IV
			80	100	22	13	>20		NR	T2-101 21.84	+149.50	21.84			III
			80	84			13.3			T2-101	+149.05 +148.91	22.29 22.43			V III
			80	100	100	100				T2-101 22.73	+148.61	22.73			II
			80	100	100	100				T2-101 24.25					
			80	100	100	100			0.4	T2-101 25.73					
			80	100	100	100				T2-101 27.24					
			80	100	100	100				T2-101 27.99	+143.35	27.99			
29													End of hole at 27.99 m.		

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

20.12.2022

CHECKED

R. Chu

DATE

21.12.2022

REMARKS



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 6

SHEET

1

of

5

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD ROTARY

CO-ORDINATES

TASK ORDER NO.

GE/2022/08.11B

MACHINE

SD44

E 829002.50

N 827546.00

DATE

05.01.2023

to

12.01.2023

FLUSHING MEDIUM

WATER

ORIENTATION VERTICAL

GROUND LEVEL

+187.35 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
05.01.2023	SW													Grey, CONCRETE. (PAVEMENT)
1									1	0.00	+187.35	0.00		Brownish yellow (10YR 6/8), clayey sandy SILT with some angular to subangular fine to medium gravel. (COLLUVIUM)
2									2	0.45	+187.25	0.10		
3									3	0.95				
4									4	1.45				
5									5	1.95	+185.35	2.00		Extremely weak, brownish yellow (10YR 6/8) dappled light red, completely decomposed, TUFF BRECCIA. (Stiff, clayey SILT)
6			80	100					6	2.00				
7	SW 3.10m							2.2 2.2,3,3 N=10	7	3.00				
8	PW								8	3.10				
9									9	3.20				
10									10	3.50				
11									11	4.00	+183.35	4.00		Extremely weak, brownish yellow (10YR 6/8) streaked grey, completely decomposed, TUFF BRECCIA. (Very stiff, slightly sandy clayey SILT with occasional angular fine gravel)
12									12	5.00				
13								3.4 4.4,4.4 N=16	13	5.10				
14									14	5.20				
15									15	5.50				
16									16	6.00				
17									17	6.00				
18									18	7.00	+180.25	7.10		Extremely weak to very weak, brownish yellow (10YR 6/8) dappled dark brownish grey, completely decomposed, TUFF BRECCIA. (Slightly clayey silty fine to coarse SAND)
19								3.5 5.6,7.9 N=27	19	7.10				
20									20	7.50				
21									21	8.00	+179.35	8.00		Extremely weak, light red (2.5YR 7/8) streaked dark grey, completely decomposed, TUFF BRECCIA. (Very stiff, slightly sandy clayey SILT)
22									22	8.00				
23									23	9.00	+178.25	9.10		Extremely weak to very weak, light red (2.5YR 7/6) mottled reddish yellow, completely decomposed, tuffaceous SILTSTONE. (Very stiff, slightly sandy clayey SILT)
24								2.4 5.8,12.15 N=40	24	9.20				
25									25	9.50				
26									26	10.00	+177.35	10.00		

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

13.01.2023

CHECKED

R. Chu

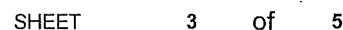
DATE

16.01.2023

REMARKS

- An inspection pit was excavated to 2.00m deep by hand tools.
- Constant head permeability tests were carried out at sections from 5.60m to 7.10m and from 11.50m to 13.00m.
- A standpipe was installed to 2.00m.
- A piezometer was installed with tip at 35.00m.

		DRILLHOLE RECORD		HOLE NO. DH 6										
		CONTRACT NO. GE/2022/08		SHEET 2 of 5										
PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk														
METHOD ROTARY			CO-ORDINATES E 829002.50 N 827546.00		TASK ORDER NO. GE/2022/08.11B									
MACHINE SD44					DATE 05.01.2023 to 12.01.2023									
FLUSHING MEDIUM WATER			ORIENTATION VERTICAL		GROUND LEVEL +187.35 mPD									
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
11	HW		70	100					21	10.00			IV	Weak, brownish yellow (10YR 6/8) dappled light grey, highly decomposed, TUFF BRECCIA. (Angular, slightly silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
									22	11.00				
									23	11.19				
12			60	73					24	12.00	+175.35	12.00	V	Very weak, reddish grey (2.5YR 7/1) mottled light red, completely decomposed, TUFF BRECCIA. (Very stiff, slightly clayey sandy SILT)
									25	12.90	+174.35	13.00		
14		0.77 at 1300 1.46 at 0800	80	83	20	0	>20		T2-101		+173.35	14.00	III	Moderately strong, light grey mottled light red and yellowish brown, moderately decomposed, TUFF BRECCIA. (CORESTONE) From 13.26m to 13.70m and 13.90m to 14.00m: Silicified.
							NR				+173.15	14.20	V	
15			80	0									IV	From 14.00m to 14.20m: No recovery, assumed to be completely decomposed TUFF BRECCIA. Weak, light red (2.5YR 7/6) and light grey (5Y 7/1), highly decomposed, silicified, TUFF BRECCIA. (Angular, slightly sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
									26	15.00	+172.25	15.10		
16			80	83	29	14	16.7		T2-101		+171.59	15.40	III	Moderately strong, light olive grey and light grey, moderately decomposed, TUFF BRECCIA. (CORESTONE)
							>20				+171.45	15.90	V	
17			80	88	0	0	>20		T2-101				III	From 15.76m to 15.90m and 16.85m to 17.00m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
							NR				+170.50	16.85	V	
18			50	0									IV	Weak, brownish yellow (10YR 6/8) mottled light grey, highly decomposed, TUFF BRECCIA. (Angular, slightly sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
									27	17.74	+169.51	17.84		
19			50	85	0	0	>20		T2-101				III	Moderately weak to moderately strong, yellow mottled light grey, moderately decomposed, TUFF BRECCIA. (CORESTONE) From 19.10m to 19.20m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
			0	80	24	24					+168.25	19.10	V	
		4.32 at 1800	0	0									IV	Weak, light olive grey (5Y 6/2) and brownish yellow (10YR 6/8) mottled light grey, highly decomposed, silicified, TUFF BRECCIA. (Angular, slightly clayey silty sandy fine to coarse GRAVEL of highly
									28	19.90	+167.35	20.00		
SMALL DISTURBED SAMPLE LARGE DISTURBED SAMPLE U76 SAMPLE PISTON SAMPLE (76mm) MAZIER SAMPLE SPT LINER SAMPLE WATER SAMPLE U100 SAMPLE STANDARD PENETRATION TEST IN-SITU VANE SHEAR TEST PACKER TEST PERMEABILITY TEST PRESSUREMETER TEST BOREHOLE TELEVIEWER PIEZOMETER TIP STANDPIPE TIP									LOGGED <u>C. Chan</u> DATE <u>13.01.2023</u> CHECKED <u>R. Chu</u> DATE <u>16.01.2023</u>		REMARKS			



DRILTECH		DRILLHOLE RECORD CONTRACT NO. GE/2022/08		HOLE NO. DH 6 SHEET 4 of 5														
PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk																		
METHOD		ROTARY		CO-ORDINATES														
MACHINE		SD44		E 829002.50 N 827546.00														
FLUSHING MEDIUM		WATER		ORIENTATION VERTICAL														
TASK ORDER NO.		GE/2022/08.11B		DATE 05.01.2023 to 12.01.2023														
GROUND LEVEL		+187.35 mPD																
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description				
31			70	100	50	30	9.4		T2-101		30.05		III	From 33.25m to 33.40m: No recovery, assumed to be completely decomposed TUFF BRECCIA.				
									T2-101		30.48							
									T2-101		30.69							
									T2-101		31.18							
32			70	100	14	8	>20		T2-101									
									T2-101		32.33		IV					
									T2-101		32.47		III					
									T2-101		32.92							
									T2-101		33.20		IV					
									T2-101		33.40		V					
34			60	81	15	0	NI		T2-101		33.70		III	Strong to very strong, dark grey and greenish grey, slightly decomposed, TUFF BRECCIA. Joints are very closely to closely spaced, occasionally medium spaced, rough planar and rough stepped, iron and manganese oxide stained, chlorite coated, dipping at 0° to 10°, 10° to 20°, 40° to 50°, 55° to 65° and subvertically from 40.30m to 40.35m. From 36.59m to 37.05m and 37.26m to 37.70m: Moderately strong to strong and moderately decomposed. From 39.05m to 40.30m and 40.60m to 41.76m: With hydrothermal alteration.				
									T2-101		34.30							
									T2-101		34.65							
35			60	100	63	23	13.3		T2-101									
									T2-101		35.45		IV					
									T2-101		35.70							
36			60	100	33	22	NI		T2-101		36.59		III					
									T2-101		36.94		II					
									T2-101		37.05		III					
									T2-101		37.26		II					
37			60	100	63	47	9.2		T2-101		37.40			From 36.59m to 37.05m and 37.26m to 37.70m: Moderately strong to strong and moderately decomposed. From 39.05m to 40.30m and 40.60m to 41.76m: With hydrothermal alteration.				
									T2-101		38.91							
									T2-101		39.87							
									T2-101		40.00							
40			60	100	100	63			T2-101		40.35							
SMALL DISTURBED SAMPLE LARGE DISTURBED SAMPLE U76 SAMPLE PISTON SAMPLE (76mm) MAZIER SAMPLE SPT LINER SAMPLE WATER SAMPLE U100 SAMPLE STANDARD PENETRATION TEST IN-SITU VANE SHEAR TEST PACKER TEST PERMEABILITY TEST PRESSUREMETER TEST BOREHOLE TELEVIEWER PIEZOMETER TIP STANDPIPE TIP									LOGGED C. Chan		DATE 13.01.2023		CHECKED R. Chu		DATE 16.01.2023		REMARKS	

										DRILLHOLE RECORD CONTRACT NO. GE/2022/08										HOLE NO. DH 6 SHEET 5 of 5									
PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk																													
METHOD ROTARY										CO-ORDINATES E 829002.50 N 827546.00										TASK ORDER NO. GE/2022/08.11B									
MACHINE SD44																				DATE 05.01.2023 to 12.01.2023									
FLUSHING MEDIUM WATER										ORIENTATION VERTICAL										GROUND LEVEL +187.35 mPD									
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description															
41		20.82 at 1800	60	100	100	80	8.0 1.0		T2-101 40.30 T2-101		40.75 41.76		II	As sheet of 4 of 5.															
12.01.2023										+145.59				End of hole at 41.76 m.															
42																													
43																													
44																													
45																													
46																													
47																													
48																													
49																													
50																													
LOGGED <u>C. Chan</u> DATE <u>13.01.2023</u> CHECKED <u>R. Chu</u> DATE <u>16.01.2023</u>										REMARKS																			